
On Using Voice Analysis in Marketing Research

Author(s): Nancy J. Nighswonger and Claude R. Martin, Jr.

Source: *Journal of Marketing Research*, Aug., 1981, Vol. 18, No. 3 (Aug., 1981), pp. 350-355

Published by: Sage Publications, Inc. on behalf of American Marketing Association

Stable URL: <https://www.jstor.org/stable/3150975>

REFERENCES

Linked references are available on JSTOR for this article:

https://www.jstor.org/stable/3150975?seq=1&cid=pdf-reference#references_tab_contents

You may need to log in to JSTOR to access the linked references.

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

American Marketing Association and Sage Publications, Inc. are collaborating with JSTOR to digitize, preserve and extend access to *Journal of Marketing Research*

Voice analysis has received attention recently both in the marketing/advertising literature and among marketing research practitioners. The authors summarize the concept behind the technique and identify major concerns with its use. They draw upon research from other disciplines, particularly psycholinguistics, psychosomatic medicine, polygraphy, and acoustics for their discussion.

On Using Voice Analysis in Marketing Research

Attitudes are constructs which exist in the minds of individuals. Marketers long have been concerned with their ability to document attitudes, and have developed numerous attitude scaling techniques in an attempt to define operationally the measurement of consumer attitudes. The most commonly used tool of attitude measurement is the self-report method (Kinnear and Taylor 1979), whereby respondents report their beliefs or feelings (either directly or indirectly) and their answers are applied to some predetermined attitude scale. However, two major problems are common with this type of measurement. First, the self-report method depends on the subject's ability and/or willingness to provide accurate information. Second, though the self-report method may tap the affective component of a subject's response, the degree of emotional commitment behind the reported attitude can be measured only by using some sort of ordinal scaling, such as a Likert scale or a semantic differential scale. This type of measurement lacks objectivity, because the researcher has no way of ensuring that each point on the scale represents the same thing for each respondent.

Physiological measurements of beliefs and feelings, in contrast, operate on the premise that we express with our bodies that which we feel in our minds. Emotional changes resulting from external or internal stimuli are believed to cause various physiological

reactions, such as changes in blood pressure, heart rate, pupillary dilation, or rate of perspiration. Because these bodily reactions are involuntary, the subject has no chance to distort the response deliberately. Although physiological measurements can objectively register a subject's emotional responses to various stimuli, a major limitation to the physiological response approach in marketing applications has been that it measures only the intensity of feelings and not the direction (positive or negative). Another difficulty with most instruments for measuring or recording physiological functions is that they can be used only while the subject sits quietly in an office or laboratory. Blood pressure, respiration and pulse rate, the psychogalvanic reflex, etc., cannot be recorded while a person is engaged in the normal activities of life. The laboratory situations, besides being artificial, are time consuming and expensive. These limitations have inhibited the use of most physiological measurements in marketing research, despite their ability to record objectively the intensity of a subject's emotional responses.

Because it is purported to overcome these obstacles, voice pitch analysis has drawn the attention of marketing researchers (Brickman 1976, 1980; Haas 1979; Nelson and Schwartz 1979). Voice pitch analysis examines changes in the relative vibration frequency of the human voice to measure emotion. As the measurement of the pitch changes requires only a tape-recorded vocal response, no complicated or cumbersome equipment is needed at the interview site. Also, because the recording of the physical phenomenon (voice pitch) occurs simultaneously with the subject's conscious interpretation of the attitude (verbal response), the direction (positive or negative) of

*Nancy J. Nighswonger is a doctoral student in Marketing, Graduate School of Business, University of Pittsburgh. Claude R. Martin is Isadore and Leon Winkelman Professor of Marketing, Graduate School of Business Administration, University of Michigan.

the attitude is ascertained from the subject's self-report and the intensity of the emotion is measured at the same time by mechanical means. In response to the current popularity of voice pitch analysis, the purpose of our article is twofold: (1) to discuss some of the technical and methodological problems involved in applying the technique to marketing situations and (2) to examine the potential of voice pitch analysis in marketing research.

APPLICATIONS OF THE TECHNIQUE

The analysis of voice patterns has been used in many different disciplines for many different purposes. Perhaps the best known application of voice analysis is in lie detection (see, for example, Kubis 1974), but it has been widely used in other fields as well. Voice analysis has been applied in psychosomatic research to measure the influence of noise on emotional states (Mason 1969). It has been used in psycholinguistics to examine the production of sound (Cairns and Cairns 1975; Lieberman 1967; Lieberman and Michaels 1962; Ostwald 1963), and in speech and music to monitor voice and articulation (Bradley 1916; Fairbanks 1960). Russian researchers have attempted to use voice analysis to monitor the stress levels of cosmonauts during space flights (Simonov and Frolov 1973). Voice analysis techniques also have been used in studies on stage fright, stress among dental patients, and diagnosis of alcoholics (see Holden 1975). Though the uses of voice analysis are diverse, they are based on the same underlying concept: a baseline pitch level is determined for an individual, and deviations from the baseline are examined and interpreted in different contexts.

THE CONCEPT

A person's body normally functions at a certain physiological pace, be it fast, slow, or somewhere between. For each individual, this normal pace is referred to as the baseline. In voice analysis, the normal or baseline pitch of an individual's speaking voice is charted by engaging the respondent in unemotive conversation. A deviation from the baseline level indicates that the respondent has reacted to a stimulus, such as a question (Ferguson and Miller 1973, p. 145).

Studies have shown the acoustic characteristics of speech to be among the most sensitive indicators of emotion (Williams and Stevens 1972, p. 1239). The analysis of the voice is an analysis of the speech signal that reflects vocal cord activity, namely, the waveform of the pulses from the vocal cords as well as the frequency of those pulses. This specific vocal cord activity is called the voice fundamental frequency. The fundamental frequency can undergo variations that may not be intended or under the overt control of the speaker.

These variations are explained biologically by the complex role of the sympathetic division of the au-

tonomic nervous system in bodily functions. Although other physiological effects can influence various components of the speech system, it has been established that the speech signal reflecting vocal cord activity is likely to be influenced by physiological changes brought about by the emotional state of the speaker (Williams and Stevens 1972, p. 1239).

Special audio-adapted computer equipment is used in voice analysis techniques to measure abnormal frequencies in the voice caused by changes in the autonomic nervous system (Brickman 1976, p. 44). For each individual these abnormal frequencies, as well as the intensity of the frequencies, are compared with the normal frequency range to obtain a net reaction "score" caused by the stimulus in question. The score is expressed as a percentage of the mean frequency for that subject. The percentage is recorded as either positive or negative, depending on whether the subject answered yes or no to the question posed. The greater the (absolute) percentage change, the greater is said to be the emotional intensity of the subject's reaction.

By this scoring system, voice analysis is claimed to identify or "filter out" those respondents who show a significant voice response (emotional reaction) to the attitudes they express in their answers. Furthermore, because of the interplay between the voice analysis and the respondent's subjective verbal reaction to a question, the direction (positive or negative) of the recorded attitude is also assessed.

Briefly, it is recognized that adults tend to modulate their voices, unconsciously, in sympathy with the emotional meanings of words (Mason 1969, p. 278). These changes in the voice may not be clearly discernable to the human ear, but when the voice inflections are plotted in subsequent computer analysis of the tape-recorded voice, even subtle changes in pitch can be detected. Voice pitch analysis is an attempt to examine the speech signals that reflect these unconscious voice patterns, because these speech signals are likely to be influenced by bodily changes resulting from the emotional state of the speaker.

A CAUTION

The voice pitch method is not as straightforward as it might seem. Many problems must be resolved in the manner in which the voice is recorded, the topics or questions used in establishing the baseline level for each respondent, and the subsequent interpretation of the noted reactions. For example, there may be individual differences in response patterns; that is, the emotional "style" of the individual may bias the voice analysis (Atkinson 1976; Hensel 1971). Past or present health conditions, such as asthma, may intensify the respondent's physiological reactions to emotional stimuli (Hahn 1966; Mason 1969) because of the possible malfunctioning of autonomic feedback mechanisms. Finally, although research is being con-

ducted, the exact physiological relationships that cause the changes in voice fundamental frequency have not been clearly determined (Atkinson 1978). Consequently, the ability of voice analysis to measure what it is purported to measure is still under scrutiny.

VOICE ANALYSIS IN MARKETING RESEARCH

In the literature on voice analysis in marketing, the use of the technique has been demonstrated in several applications. Brickman (1976, 1980) used the technique in three ways: (1) in package research, (2) to predict consumer brand preference for dog food, and (3) to determine which consumers from a target group would be most predisposed to try a new product. Nelson and Schwartz (1979) applied voice analysis to measure consumers' emotional responses to advertising, and compared the predictive power of the voice response scale with that of the 10-point attitude scale. In all of these studies, the predictions made from voice analysis were found to reflect actual results better than predictions based on more conventional methods of questioning. However, making such predictions without being able to explain the reason for their greater accuracy is not theoretically sound. The observed relationships may be spurious, in which case the technique would be masking rather than uncovering the variable(s) critical to the research.

Marketing research, advertising, and other firms that employ voice analysis in marketing contexts have neglected some of the fundamental issues in the application of the technique. These issues can be divided into two major categories: technical concerns, pertaining to the feasibility of voice analysis itself, and methodological concerns, pertaining to some characteristics of voice analysis that must be accommodated in application of the technique to marketing problems.

Technical Concerns

One of the claimed advantages of voice analysis in marketing is the ability of the technique to filter out "lip service" responses from consumers. A lip service response is defined as a response that is something less than a committed response; the subject could be giving the "socially acceptable" response, a confused response, or a response that will merely suffice to fulfill the obligation to the questioner. Voice analysis procedures filter out these so-called lip service responses, and identify those subjects who show an emotional commitment (a significant voice pitch change) to the attitude reported to the interviewer. According to VOPAN Marketing Research of Boston (VOPAN 1980, p. 4), in most cases outside stimulations or internal thoughts do not cause emotions, and subjects in those cases will express attitudes that are in the category of lip service responses. It is also important to note that if the respondent is unduly confused or lying (either consciously or unconscious-

ly), the recorded pitch change in the responses will be extremely large. These responses are also filtered out of the system, although Nelson and Schwartz (1979) report that no more than 5% of their samples had shown such extreme voice changes. Eliminated, too, are the subjects whose widely fluctuating vocal characteristics indicate that they feel uncomfortable in the interview situation (Brickman 1976, p. 45).

Because of the process of filtering out unemotional responses, voice analysis has the potential for determining which subjects are emotionally moved by a message and, hence, are likely to act on it. Theoretically the logic is appealing, but in the application of voice analysis this concept of emotional commitment must be operationalized by placing limits on what constitutes a significant voice pitch change. Operationalizing "emotional commitment" poses a major problem, because it has not been established that individuals are consistent in their physiological responses to emotion. Whether an individual's voice will exhibit the same reaction to the same emotion at different points in time is unknown. Also unknown are the effects of different emotions on a given speaker's voice, and whether the same emotions produce the same voice changes in different people. It is not clear, then, that the filtering process achieves in practice what it is claimed theoretically to achieve. The cutoff point for what constitutes an emotional commitment is arbitrary; consequently, so is the assignment of individuals to the categories of "emotionally committed" or "not emotionally committed." Given that there may be individual differences in physiological response patterns, applied voice analysis cannot identify with certainty the specific reason(s) for a particular subject's inclusion in or exclusion from either category. In other words, the process of identifying lip service responses has no theoretical substantiation. This claimed ability of voice analysis has not been tested in validation studies; rather, the researchers have assumed that the responses identified by the technique as lip service responses actually are such responses. More specifically, the researchers have neglected to examine the ability of the voice analysis scale to categorize a subject's response appropriately.

For illustration, remember that each individual has a normal or baseline speaking voice as well as a range within which his or her voice can fluctuate. An emotional response is reflected physiologically by a change in the level of voice pitch, and the direction (positive or negative) of that response is determined by the respondent's subjective evaluation of the emotion. The first scale of measurement used in voice analysis identifies relative pitch changes, not absolute pitch levels. When interpreting the voice pitch measurements, however, the researcher examines not only the degree of change in pitch level (intensity of emotion), but also the direction of the response (posi-

tive or negative emotion). Hence, the ultimate scale employed is intended to show intensity as well as direction of the recorded emotion. It is a continuous scale that ranges from intense reactions in a negative direction (a "no" response and an extreme voice pitch change) to intense reactions in a positive direction (a "yes" response and an extreme voice pitch change).

To date, marketing researchers have used only the positive end of this voice pitch scale, that is, those subjects who respond "yes" to a trial interest question and concurrently register a heightening of voice pitch within a specified range. Those subjects who register an emotionally charged "no" response, an extreme response at either end of the scale, or an insignificant (lip service) response are eliminated from further analysis. This practice does not test the capabilities of voice analysis. Rather than ignoring the subjects who do not score on the positive end of the scale, researchers should explore their attitudes by some other form of measurement. For example, how many of the subjects who are eliminated from, say, a study in new product research actually later purchase the product? It seems appropriate, in addition to determining how many purchases voice analysis *does* predict, to determine how many purchases the technique *does not* predict. The entire range of possible responses must be evaluated and tested for its actual correspondence with consumer behavior before the validity of the technique can be established.

The voice analysis filtering process poses yet another problem for the practical application of the voice analysis technique. As it has been defined, the filtering process eliminates from further analysis those subjects who are confused, lying, or uncomfortable, along with those who are said to be truly uncommitted in their responses. If voice analysis actually has the capability of making these distinctions, the kinds of responses the voice analysis technique can utilize appear to be limited. It follows that the kinds of topics that can be used with voice analysis are also limited. A topic that is risky or complicated may provoke a confused response; a sensitive topic may cause the subject to feel uncomfortable and/or to lie with a socially acceptable answer; an emotional topic may cause extreme emotional involvement and, accordingly, extreme voice pitch change; and an unemotional topic may induce no significant voice pitch change from the respondents. What the voice analysis technique *cannot* measure seems clearly defined, but what the technique actually *can* gauge—at least in a marketing sense—is vague.

Voice analysis must develop in technical sophistication before it can be applied with confidence in marketing. Specifically, to explain the apparent accuracy of the technique, we need to know more about the nature of the voice as it responds to emotion. The scale used in measuring voice fundamental frequencies then could be refined so as to distinguish

more clearly between the different "levels" of voice response (confused, lying, etc.).

Methodological Concerns

Two aspects of studies using the voice analysis technique are of methodological concern in terms of both the research design used with the technique and the interpretation of the results. Specifically, these two aspects are the method of questioning and the absence of a control group, both of which can substantially affect the results of an experiment.

Method of questioning. Evidence indicates that once an emotional response has been registered in the voice, the pitch of the voice does not return immediately to the normal baseline level (Dunbar 1954; Friedhoff, Alpert, and Kurtzberg 1973). If neutral (unemotive) questions are not included between the questions designed to elicit an emotional reaction, the control baseline is not reestablished after each emotional response that is registered and, hence, it cannot be used as a neutral comparison across all responses. If the subjects are given a battery of questions designed to elicit emotional reactions with no neutral questions between (as apparently has been done by marketers) a static formula for calculating deviations in pitch from the mean level is impossible to apply accurately.

Absence of a control group. The validation studies that have been done have not included controls against which to evaluate the performance of the voice analysis measures. The effects of this lack of control can be seen in two examples.

First, in a study predicting consumer brand preference for dog food (Brickman 1976), all subjects were identified as using one of six brands of dog food on a habitual basis, but they were instructed not to reveal which of the six brands they actually used. The voice analysis technique was then employed to predict each consumer's actual brand preference. However, there were no "control" subjects who used none of the six brands on a habitual basis. Consequently (and possibly aggravated by the fact that the subjects were aware of the purpose of the study), the attributes that were determined in the first interview to constitute the subject's "ideal brand" may, in fact, have come subjectively very close to describing the brand the customer actually used. If the subject were already a habitual user of a particular dog food, the purchase decision process would have been reduced to an almost automatic behavior. The degree of familiarity with six different brands of dog food, then, appears to be limited. The subject's frame of reference for answering the attribute questions would be restricted to past experience, which theoretically reduces his or her range of knowledge about dog food to one or two brands because the subject has been defined as a habitual user of one particular brand. The correct prediction of the subject's actual brand seems almost inevitable. If familiarity with a topic affects the sub-

ject's attitude or commitment toward that topic, the results of the study may have been seriously biased by the lack of control subjects.

The second example illustrates a problem in advertising research, although the same effects can be generalized to almost any area of marketing. Here, the lack of a control refers to the lack of a reference "zero point," that is, an objective standard by which to evaluate the voice analysis scores produced for each element in the experiment. Three commercials that had been pretested in test markets were also tested by voice analysis techniques to determine which commercial was most effective in generating sales (VOPAN 1980, p. 19). Though the voice analysis scores matched the test market data in terms of the effectiveness rankings of the three commercials (commercial X had a lower voice analysis score than Y, which had a lower score than Z, where Z was clearly the most effective commercial in the test market), the voice analysis scores indicated nothing about the share of market that was expected to be generated by the three commercials. There was no zero point, no "choose none of the three" category. It is possible that, of the three commercials, none was expected to generate enough sales to justify the advertising expenditure. Market-test data could indicate a problem of this type, whereas voice analysis data would not. A ranking of the available choices can be obtained from the voice analysis scores, but these scores are only ordinal. In other words, one limitation of the voice analysis procedure is that it will not provide information about whether or not all of the alternatives are good choices, or whether or not they are all poor choices. This problem is common, however, especially with new scales of measurement. Experience with the new technique is required before the scale can be calibrated. Voice analysis at its present stage is limited to a relative evaluation of alternatives.

IMPLICATIONS FOR CURRENT RESEARCH

Voice analysis depends on physiological changes in the body that occur as a result of emotion. Ordinarily, emotion does change bodily states and quantitative relationships in the body, but there is no evidence to indicate that a particular bodily response is associated with any particular emotion. Thus, any given element in the complex network of bodily reactions may or may not change with emotional states (Dunbar 1954).

This variability in physiological responses may be a function of personal style (Hensel 1971; Williams and Stevens 1972) and/or of the specific emotion being felt by the respondent (Ferguson and Miller 1973, p. 157). Whatever the source of this variability, one physiological measure, standing alone, is not sensitive to every emotion nor is it sensitive to any particular set of emotions for every individual. This evidence implies that voice analysis should be used in conjunc-

tion with, rather than in place of, other methods of attitude measurement.

In the absence of conclusive research on the details of the voice as a physiological correlate of emotion, we hypothesize—for the time being—that voice analysis does, in fact, measure the direction (positive or negative) of an attitude as well as the emotional commitment behind that attitude. On the basis of this hypothesis, immediate research efforts should focus on establishing the concurrent validity of voice analysis as an attitude measure. Many tests to explore this concurrent validity are potentially possible. However, there appears to be a logical link between voice pitch analysis and response latency research (Aaker et al. 1980; MacLachlan, Czepiel, and LaBarbera 1979).

Response latency is defined as the amount of time a respondent spends in deliberation before answering a question. MacLachlan et al. (1979) found that response latency is a robust measure that serves as an indicator of the subject's certainty in a given response. Thus, response latency has been shown to be a valid and sensitive measure of the conviction with which attitudes are held. Because data used in voice analysis are recorded on tape, the response latency of the subjects can be easily and inexpensively obtained along with the voice pitch measurements. Combining voice and response latency analyses offers the promise of simultaneously measuring the same phenomenon and, thus, testing for concurrent validity. This linkage of the two methods is also the basis for the next research step.

Aside from the test for concurrent validity, combining voice pitch analysis with response latency research may answer our previous criticism of the use of a single physiological measure. In other words, voice analysis combined with the response latency technique may be a more sensitive and effective measure of attitudes than voice analysis alone.

POTENTIAL OF VOICE ANALYSIS IN MARKETING RESEARCH

If the technical difficulties can be overcome, voice analysis holds considerable promise in the area of marketing research.

A measure such as voice analysis that is designed to filter out lip service responses to a series of questions would undoubtedly be a boon to market segmentation research. The ability to identify those respondents who are emotionally committed to their expressed attitudes also would be useful in such areas as copy testing and commercial wearout research, especially if the emotional content of the voice were to be monitored over repeated exposures to a particular advertisement.

The voice analysis "filtering" process can also be applied from the opposite direction, that is, by using the technique to filter out those respondents who are emotionally committed to their attitudes, and concen-

trating further research efforts on those respondents who did *not* show an emotional commitment in their answers. To the marketing practitioner desiring to increase primary demand and/or market share for the company's product, the *lack* of emotional involvement could be more interesting than the presence of an emotional commitment. Certainly researchers involved with attitude structure and attitude change could also benefit from these voice analysis methods.

The list of potential applications of the technique is almost limitless. There is a real danger, however, in becoming carried away with the theoretical potential of a research instrument before that instrument has validated its abilities in practice. At this stage in the development of voice analysis we cannot say whether the technical difficulties can be overcome. Neither do we know whether voice analysis methods will prove to be more effective or efficient than the techniques currently available. Clearly, further technical research is needed to define the capabilities and limitations of the technique before it can be employed with confidence in the marketing discipline.

REFERENCES

- Aaker, David A., Richard P. Bagozzi, James M. Carman, and James M. MacLachlan (1980), "On Using Response Latency to Measure Preference," *Journal of Marketing Research*, 17 (May), 237-44.
- Atkinson, James E. (1976), "Inter- and Intraspeaker Variability in Fundamental Voice Frequency," *Journal of the Acoustical Society of America*, 60 (August), 440-5.
- (1978), "Correlation Analysis of the Physiological Factors Controlling Fundamental Voice Frequency," *Journal of the Acoustical Society of America*, 63 (January), 211-22.
- Bradley, Cornelius Beach (1916), . . . *On Plotting the Inflections of the Voice*. Berkeley: University of California Press.
- Brickman, Glen A. (1976), "Voice Analysis," *Journal of Advertising Research*, 16 (June), 43-8.
- (1980), "Uses of Voice-pitch Analysis," *Journal of Advertising Research*, 20 (April), 69-73.
- Cairns, Helen and Charles Cairns (1975), *Psycholinguistics: A Cognitive View of Language*. New York: Holt, Rinehart and Winston.
- Dunbar, Flanders (1954), *Emotions and Bodily Changes*, 4th edition. New York: Columbia University Press, 158.
- Fairbanks, Grant (1960), *Voice and Articulation Drillbook*, 2nd edition. New York: Harper.
- Ferguson, Robert J. and Alan L. Miller (1973), *The Polygraph in Court*. Springfield, Illinois: Thomas Books.
- Friedhoff, Arnold J., Murray Alpert, and Richard L. Kurtzberg (1973), "An Effect of Emotion on Voice," *Nature*, 193 (4813), 357-8.
- Haas, Charlie (1979), "Charlie Haas on Advertising," *New West* (November 5), 31-59.
- Hahn, W. W. (1966), "Autonomic Responses of Asthmatic Children," *Psychosomatic Medicine*, 28, 323-32.
- Hensel, James Stephen (1971), "Physiological Measures of Advertising Effectiveness: A Theoretical and Empirical Investigation," doctoral dissertation, The Ohio State University.
- Holden, Constance (1975), "Lie Detectors: PSE Gains Audience Despite Critics' Doubts," *Science*, 190 (October 24), 359-62.
- Kinnear, Thomas C. and James R. Taylor (1979), *Marketing Research: An Applied Approach*. New York: McGraw-Hill Book Company, 301.
- Kubis, Joseph A. (1974), "Comparison of Voice Analysis and Polygraph as Lie Detection Procedures," *Polygraph*, 3 (March), 1-41.
- Lieberman, Philip (1967), *Intonation, Perception and Language*, Research Monograph No. 38. Cambridge, Massachusetts: The M.I.T. Press.
- and S. Michaels (1962), "Some Aspects of Fundamental Frequency, Envelope Amplitude, and the Emotional Content of Speech," *Journal of the Acoustical Society of America*, 34, 922-7.
- MacLachlan, James, John Czepiel, and Priscilla LaBarbera (1979), "Implementation of Response Latency Measures," *Journal of Marketing Research*, 16 (November), 573-7.
- Mason, R. K. (1969), "The Influence of Noise on Emotional States," *Journal of Psychosomatic Research*, 13, 275-82.
- Nelson, Ronald G. and David Schwartz (1979), "Voice-Pitch Analysis," *Journal of Advertising Research*, 19 (October), 55-9.
- Ostwald, Peter F. (1963), *Soundmaking: The Acoustic Communication of Emotion*, American Lecture Series No. 538. Springfield, Illinois: Thomas Books.
- Simonov, P. V. and M. V. Frolov (1973), "Utilization of Human Voice for Estimation of Man's Emotional Stress and State of Attention," *Aerospace Medicine*, 44, 256-8.
- VOPAN Marketing Research (1980), "VOPAN® Voice Pitch Analysis in Marketing and Advertising Research," unpublished company literature, The VOPAN Company, Boston (January).
- Williams, Carl E. and Kenneth N. Stevens (1972), "Emotions and Speech: Some Acoustical Correlates," *Journal of the Acoustical Society of America*, 52 (4), 1238-50.